Anisotropic Minimum Dissipation (AMD) model

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Introduction to the Model

In the field of computational fluid dynamics, one often finds oneself searching for ways to reduce computational cost while maintaining accuracy. This task becomes increasingly difficult at high Reynolds number, where large-scale eddies transfer their energy to progressively smaller scale eddies until eventually, viscosity dissipates kinetic energy as internal energy (heat). DNS allows one to numerically solve the Navier-Stokes equations accurately at all scales without any modelling assumptions; however, this is often computationally infeasible since the required grid resolution scales approximately as $Re^{9/4}$.

A fix for this is to apply a spatial filter to the Navier-Stokes equations, allowing us to solve for the velocities of large scale motion exclusively. The unresolved small scales are then represented through the implementation of a subgrid-scale (SGS) model. This is exactly the idea of Large Eddy Simulation (LES). Different LES models vary in how they treat subgrid scales, ranging from classical eddy-viscosity models to more advanced dynamical models.

In this project, we focus on implementing the **Anisotropic Minimum Dissipation (AMD)** model. This SGS model is designed to introduce only the minimum amount of dissipation to ensure numerical stability whilst preserving the physical accuracy. Unlike many typical eddy-viscosity models, AMD adapts locally to the gradients of the flow and only activates additional dissipation when resolved scales show energy transfer. This implies that on relatively-coarse grids, AMD prevents excessive artificial dissipation from suppressing key flow dynamics.

To analyse the AMD model, we apply it to **Taylor-Green vortices (TGV)**, a standard benchmark for assessing models of turbulent flows. TGVs begin as smooth, laminar flow with a periodic large-scale structure, evolving into turbulence through vortex stretching and eventually viscous dissipation. This makes the Taylor-Green vortex an ideal test case due to its self-generating energy cascade, ideal inertial-range behaviour and energy dissipation.

Methodology

DNS Code Modifications and LES Implementation

The starting point for this project was a pseudo-spectral DNS solver for the Taylor–Green vortex. To enable Large Eddy Simulation with the Anisotropic Minimum Dissipation (AMD) model, the solver was reformulated using a finite-difference, flux-form discretisation on a uniform grid. The model derivation follows those of Rozema, Bea, Moin and Verstappen in [1].

Finite-difference reformulation. The original spectral differentiation was replaced with finite-difference operators acting on cell-centred velocity fields. Convective and viscous fluxes were evaluated at cell faces, requiring:

- backward differences for face-centred gradients (viscous fluxes),
- forward differences for divergence at cell centres,
- second-order central differences for AMD gradient evaluation.

This face–cell arrangement allows the Navier–Stokes equations to be written in conservative form, facilitating the inclusion of subgrid-scale viscosity.

Projection method. The incompressibility constraint was enforced using a projection method. A pressure Poisson equation was solved in Fourier space using an FFT-based solver, and the resulting pressure gradient was used to project the tentative velocity field onto a divergence-free space.

AMD subgrid-scale model. The AMD model introduces a locally adaptive eddy viscosity based on the resolved velocity gradients. At each time step, the full velocity gradient tensor G_{ij} was computed using second-order central differences, from which the symmetric strain-rate tensor S_{ij} was formed. The AMD viscosity was computed as

$$\nu_{\mathrm{AMD}} = \max\left(0,\ -C_{\mathrm{AMD}}\,\Delta^2 \frac{G_{ij}S_{ij}}{G_{ij}G_{ij}+\varepsilon}\right),$$

where Δ is the filter width, C_{AMD} is the model's constant and ε is set equal to 10^{-12} to prevent potential division by 0 errors (although this is typical convention, it is generally unnecessary for turbulent flows). The tensor contraction $G_{ij}S_{ij}$ measures the local transfer of kinetic energy between resolved scales. When this quantity is negative, the resolved flow is producing subfilter-scale energy, and the exact subfilter tensor would dissipate energy. In such regions, AMD introduces the aforementioned positive eddy viscosity. [1] shows, via a Taylor expansion of the sub-filter tensor, that this eddy-viscosity model is consistent with the exact eddy dissipation of the exact sub-filter tensor. The effective viscosity used in the viscous fluxes is then

$$\nu_{\mathrm{eff}} = \frac{1}{Re} + \nu_{\mathrm{AMD}}.$$

This is interpolated to cell faces and inserted directly into the viscous fluxes. This allows the SGS dissipation to enter the momentum equation in conservative form, consistent with the original DNS solver. The convective fluxes remain unchanged, ensuring that the resolved-scale dynamics are treated identically to the DNS formulation. In this way, the solver behaves as DNS for the large scales and as LES with AMD for the small scales. This formulation for viscosity ensures that dissipation is added only when the resolved flow exhibits compressive or energy-transferring behaviour, preventing excessive damping.

Grid Resolutions and Numerical Parameters

Simulations were performed on uniform grids of size

$$16^3, \quad 32^3, \quad 64^3.$$

The computational domain was the periodic cube

$$[0, 2\pi]^3.$$

Time integration. A forward Euler scheme was used with a time step determined by a CFL condition:

$$\Delta t = \mathrm{CFL} \cdot \Delta x, \qquad \mathrm{CFL} = 0.15.$$

Flow parameters. The Reynolds number was fixed at

$$Re = 1600,$$

and the AMD constant was varied in the range $C_{\mathrm{AMD}} = 0.1$–0.5, with $C_{\mathrm{AMD}} = 0.3$ used for the primary results. The filter width Δ was set equal to the grid cell width, $\Delta = \Delta x$, which is uniform throughout the flow.

Initial condition. All simulations were initialised with the classical Taylor–Green vortex:

$$u = \cos x \sin y \sin z, \qquad v = -\sin x \cos y \sin z, \qquad w = 0,$$

which undergoes vortex stretching, transition to turbulence, and eventual decay. This makes it an ideal benchmark for assessing the performance of the AMD model against DNS.

Results

The results section will be split into the following parts:

- Energy dissipation
- Dissipation rate
- Energy spectra
- Grid refinement
- AMD constant selection
- Computational time

Unless otherwise stated, the grid will be of size 64^3, for which the original DNS code was given. As mentioned before, for most of the results we will be using the AMD constant $C_{\text{AMD}} = 0.3$. This is motivated from the research of Rozema, Bea, Moin and Verstappen [1]. In their paper chapter II.B they note that for a second order accurate model (which is the same for us as central differences are used) the constant 0.3 should be used.

We will be comparing our AMD results with results obtained from using a DNS solver on a fine grid (64^3 number of grid cells). In this report, this solution will be treated as the actual solution to this flow problem.

Sanity check - energy dissipation

Before comparing with the given DNS solver, one has to make sure that our implemented model behaves physically - it has positive dissipation.

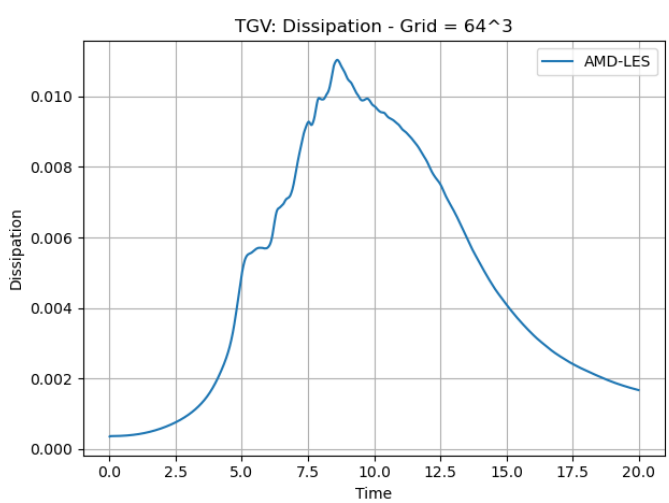

Figure 1: Energy dissipation for the AMD model.

We first note that our model indeed does have a positive dissipation. Moreover, it is consistent with literature [2]. In this paper, the authors show us dissipation shapes which match our simulations and we also get the peak of dissipation at about $t = 8$ as noted in the referenced paper.

To fully be sure about the energy dissipation of our model, we also compare it with the given DNS code.

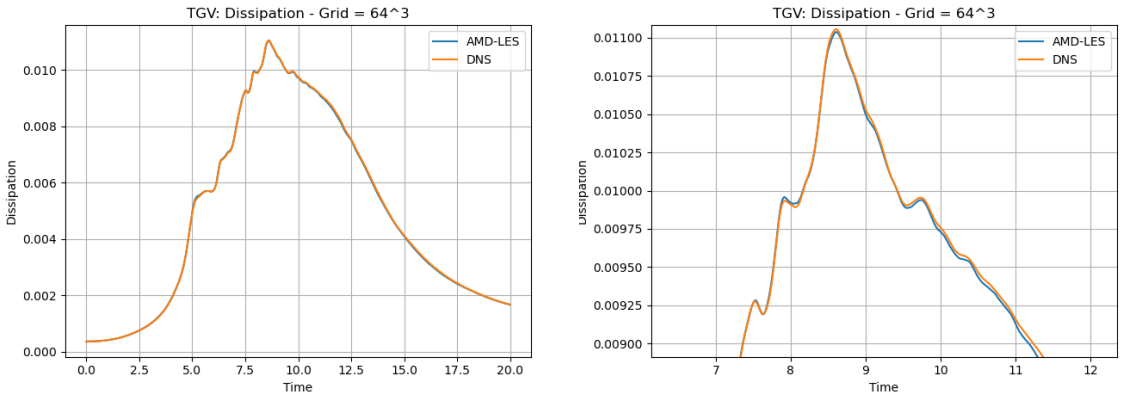

Figure 2: Energy dissipation of AMD compared to DNS

The results turn out to be incredible, as the the orange line representing the benchmarking algorithm almost overlaps with the dissipation from the AMD model (see the left picture for the entire time-domain). Only when one zooms in closer is it possible to see that the plots in fact do not coincide. The right picture shows us the energy dissipation of the 2 algorithms at the point in time where turbulence is the most active. We can note the blue(AMD) line is sitting just below the orange(DNS) line, which signals that our AMD method is slightly under-dissipative. The results are nonetheless extremely satisfactory.

Dissipation rate

For the dissipation rate we have a similar story, where the results obtained from AMD match DNS almost perfectly. Again, only when zooming in is it possible to notice that the methods in fact do produce various results. Here, at the end of the time domain, we can note that the AMD-LES is by a small fraction more dissipative than the DNS solution.

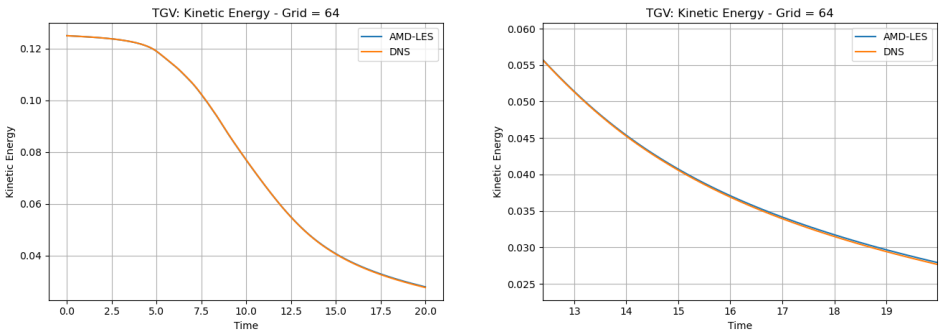

Figure 3: Energy dissipation rate of AMD compared to DNS

Energy spectra

A standard check for testing various AMD models is to look at the energy spectrum and mainly to see how the medium sizes eddies behave. By plotting the energy spectra we can see the energy for waves with various wave numbers. Again, our results do not disappoint. The line representing AMD closely matches the one for DNS. We have also included the reference line $k^{-5/3}$ which corresponds to the Kolmogorov slope. We note how our energy spectra for medium sized eddies (wave number $k \approx 10$ to 30) follows this line, which matches the theory [3].

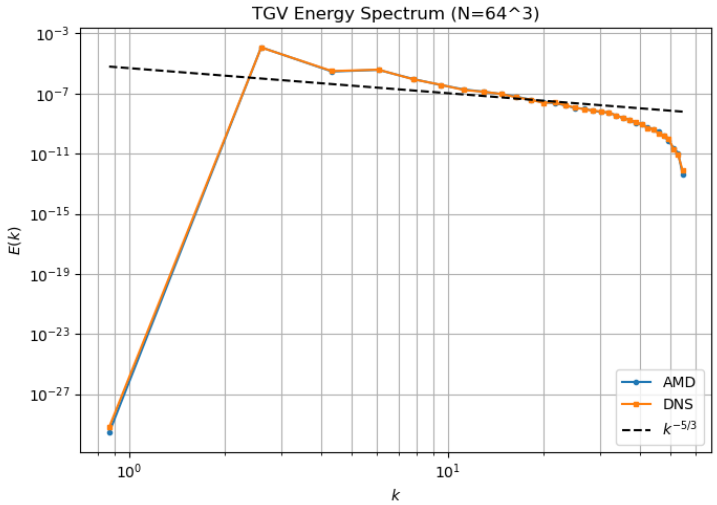

Figure 4: Energy spectra of AMD compared to DNS. Together with the Kolmogorov scale.

Grid refinement

In the grid refinement section we will take the result given to us by the DNS (with grid size 64^3) as the golden standard. We will compare this standard with LES AMD models with varying grid sizes.

Note: Computing the results using DNS on a coarse grid, does not make sense for comparisons. The DNS algorithm needs a fine grid to achieve accurate computations whereas algorithms using LES modelling can try to work around losing small scales because of the grid.

The following results for the energy dissipation, dissipation rate and energy spectra were produced.

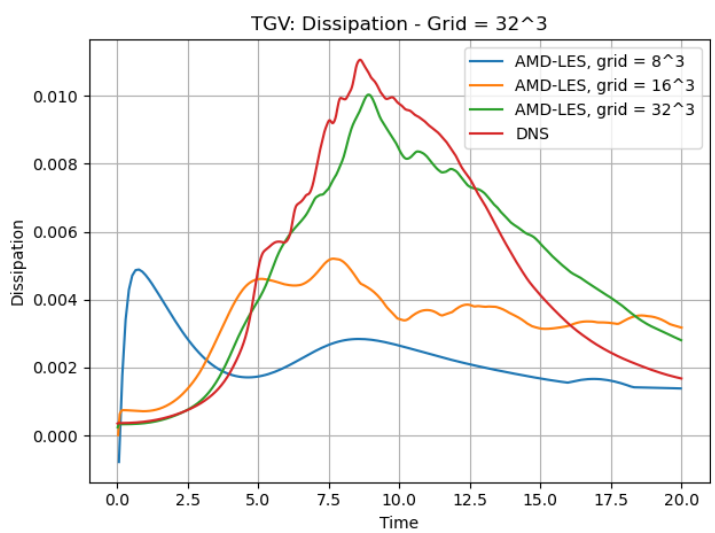

Figure 5: Energy dissipation for the AMD models with various underlying grids.

For the energy dissipation one can note that decreasing the grid cell number to 16^3 already fully breaks the structure of the desired "mountain" shape. For such a grid cell number and for 8^3, we see results that do not follow the general trends of the DNS solution.

However, when solving with AMD with grid cell number 32^3 we can still see the structural similarities of the DNS solution. We do have that the LES is under-dissipative during the peak and that it is over-dissipative after $t = 12.5$ (after the peak). However, one should still applaud the LES model as a grid with 32^3 grid cells has ~ 230000 fewer grid cells than the DNS grid.

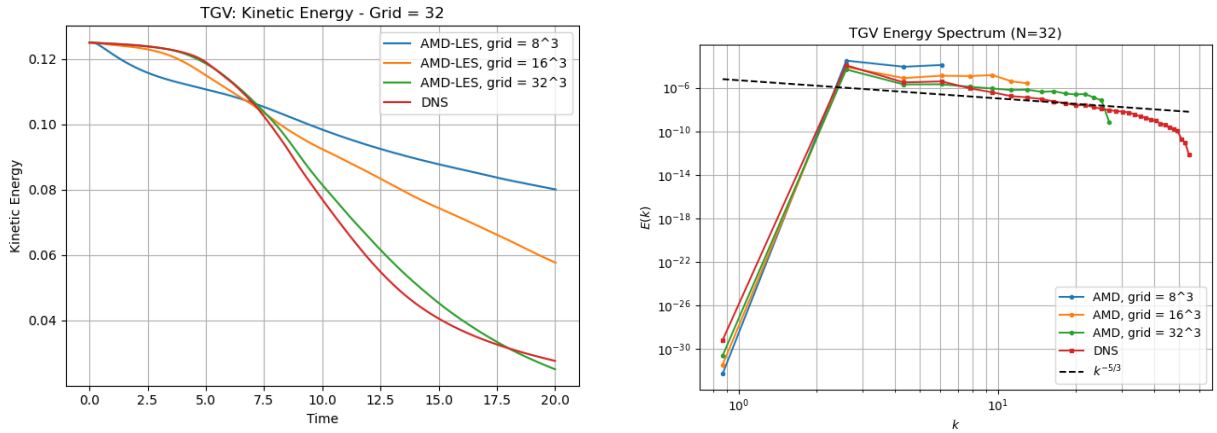

Figure 6: Energy dissipation rate and the energy spectra for the AMD models with various grids.

The figure to the left demonstrates to us that the dissipation rate of the AMD algorithms is always under-dissipative before the peak at $t = 8$ and over-dissipative after the peak. For the AMD on a grid with 16^3 grid cells we can still see the energy rate decently matching the energy rate of DNS yet after the peak it moves quite far away from the DNS refer. The green line corresponding to 32^3 grid cells does again quite well.

The energy spectrum is an interesting result. We can note how each time the grid gets coarser the spectra moves further from the reference $k^{-5/3}$ line. The medium sized scales are however quite parallel to the reference line for all of the AMD results, which notifies that the AMD model is quite good as even with such harsh grid cell numbers the medium size eddies resolve energy well.

Note: We can note that the horizontal range of the plots for each grid refinement differ in size. This is because for coarser grids small scale motion does not exist.

In conclusion, from the grid refinement study we can understand that even by changing the number of grid cells drastically, we still obtain decent results (as shown for AMD with 32^3 grid cells). When one decreases the grid cell numbers further the dissipation plots do not resemble the correct results. However, the energy spectra tells us that there is nothing wrong with our model as the medium scales get resolved as they should in theory.

Changing the AMD constant

All throughout the literature, we found that the AMD constant is typically taken as 0.3 (or in some case 1/3). We confirm this choice for C_{AMD} in this short subsection, in which we look at changing $C_{\text{AMD}} = 0.1$ and $C_{\text{AMD}} = 0.5$.

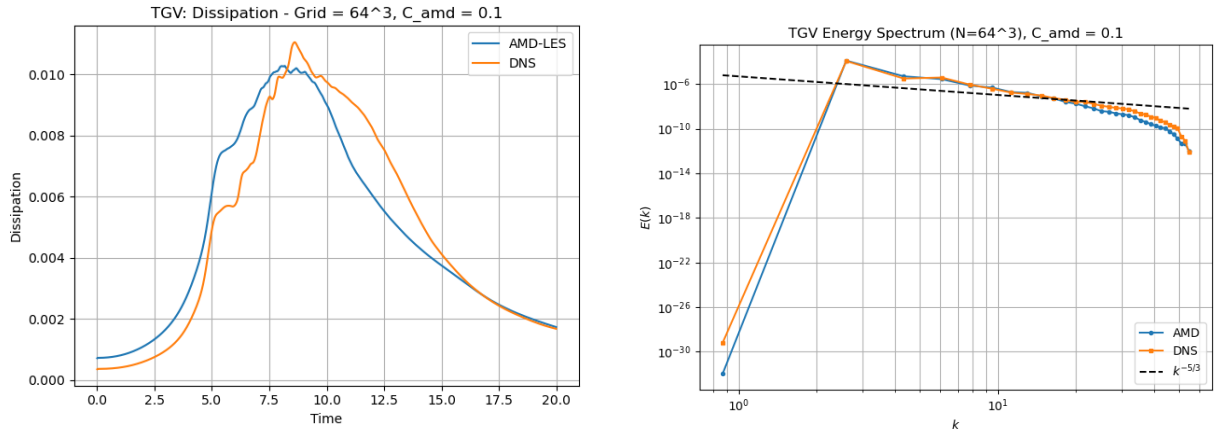

Figure 7: $C_{\text{AMD}} = 0.1$

When taking the constant too low, we note the under-correction done by the AMD model. In the first plot, we note how the model is largely over-dissipative before the peak and quite under-dissipative - after. Also, one can note the energy spectra, in which we see that because of the constant being so low, we have under-dissipation for both low and high wave-numbers.

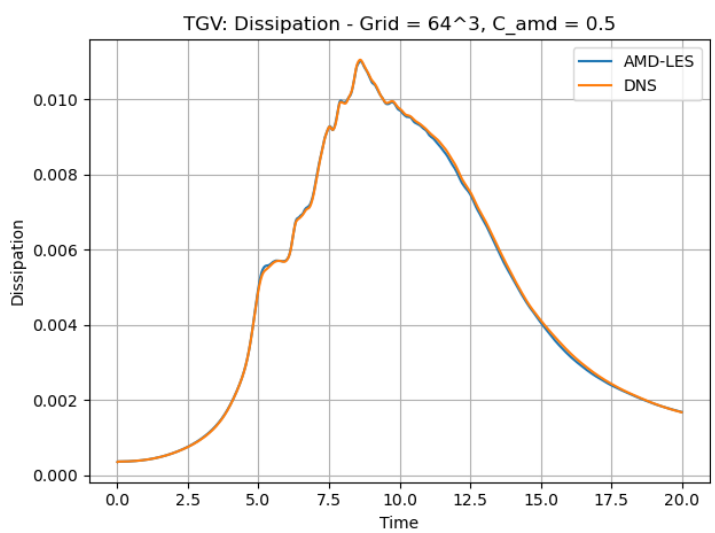

Figure 8: $C_{\text{AMD}} = 0.5$

The dissipation plot for $C_{\text{AMD}} = 0.5$ is not as bad as it was for a lower constant. However, for $C_{\text{AMD}} = 0.3$ from before, we did see less of the blue line repre-

senting AMD. More experiments should be performed to explain the fair results obtained by increasing the constant but that is outside of the scope of this report. All together, the results are in fact better with the constant $C_{\text{AMD}} = 0.3$ shown in literature, which is why for us it was chosen to be the standard.

Time

In this section we now advocate for LES when running-time is concerned.

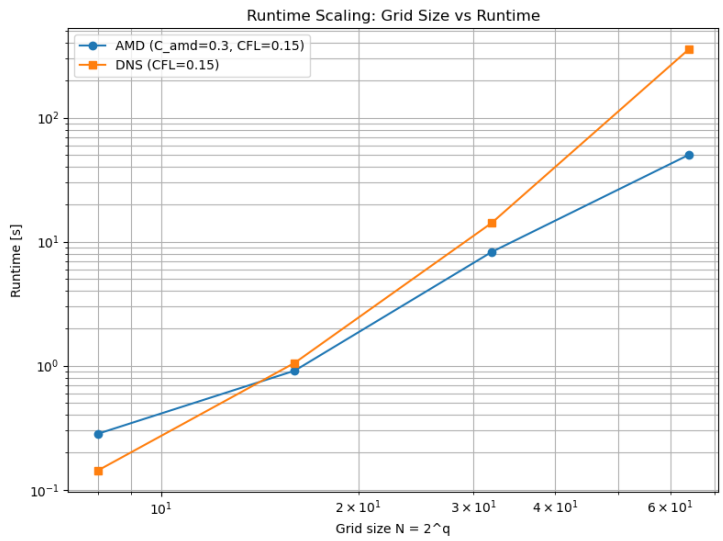

Figure 9: Time comparison: AMD vs LES

In the plot above one can note, we have demonstrated the running-time for the AMD and DNS algorithm for varying numbers of grid cells. One can note AMD outperforming DNS as the number of grid cells increases. It is even possible to see a trend for both slopes, which display that with a finer grid the difference between running times will be even larger - in favour of AMD.

Discussion of the results

In the results section we have shown various simulations using the implemented AMD code. We saw how the dissipation match costly DNS algorithms, which also held true for the energy dissipation rate and spectra. Because of such fine approximations we can conclude that our AMD implementation is in fact working well.

We also did a grid refinement study. We showed general trends when one decreases the number of grid cells the grid uses. Moreover, we demonstrated that

AMD preserves the structure of the dissipation plot even when the grid is reduced to having 32^3 number of grid cells instead of 64^3 used by DNS. In this section, we also noted how the energy spectra behaves well for all the tested grids.

We experimented a bit with the AMD constant and found that the most reliable choice is to use $C_{\mathrm{AMD}} = 0.3$ as stated in [1]. And lastly, we noted one of the strengths of AMD when compared to DNS. By using AMD we cut down on computational time as small scales are not being calculated yet modelled.

Conclusions, summary

In this report we have studied how the Anisotropic Minimum Dissipation model behaves when applied to the Taylor-Green vortices. We studied this flow using a high Reynolds number ($Re = 1600$) therefore turbulence was expected.

In the first half of this paper, we introduced the concept of Large Eddy Simulations together with explaining the AMD model. A brief introduction of the Taylor-Green vortices was given, after which AMD code implementation part began. We described the mathematics behind AMD and added the steps taken for us to change the given DNS code to suit the structure of AMD.

The second half of the report was spent on analysing various result obtained from using the AMD model. We found out that the implemented algorithm performs extremely well and that the energy spectra behaves as it should. Combining this together with the fact that AMD algorithm took less time to run, we can strongly recommend to the reader to try out AMD modelling for solving their next fluid flow problem.

References

[1] Wybe Rozema, H. Jane Bae, Parviz Moin, and Roel Verstappen. Minimum-dissipation models for large-eddy simulation. *Physics of Fluids*, 27(8):085107, 2015.

[2] James R. DeBonis. Solutions of the taylor-green vortex problem using high-resolution explicit finite difference methods. Technical Report NASA/TM—2013-218024, NASA Glenn Research Center, Cleveland, Ohio, 2013.

[3] Stephen B. Pope. *Turbulent Flows*. Cambridge University Press, Cambridge, 2000.